

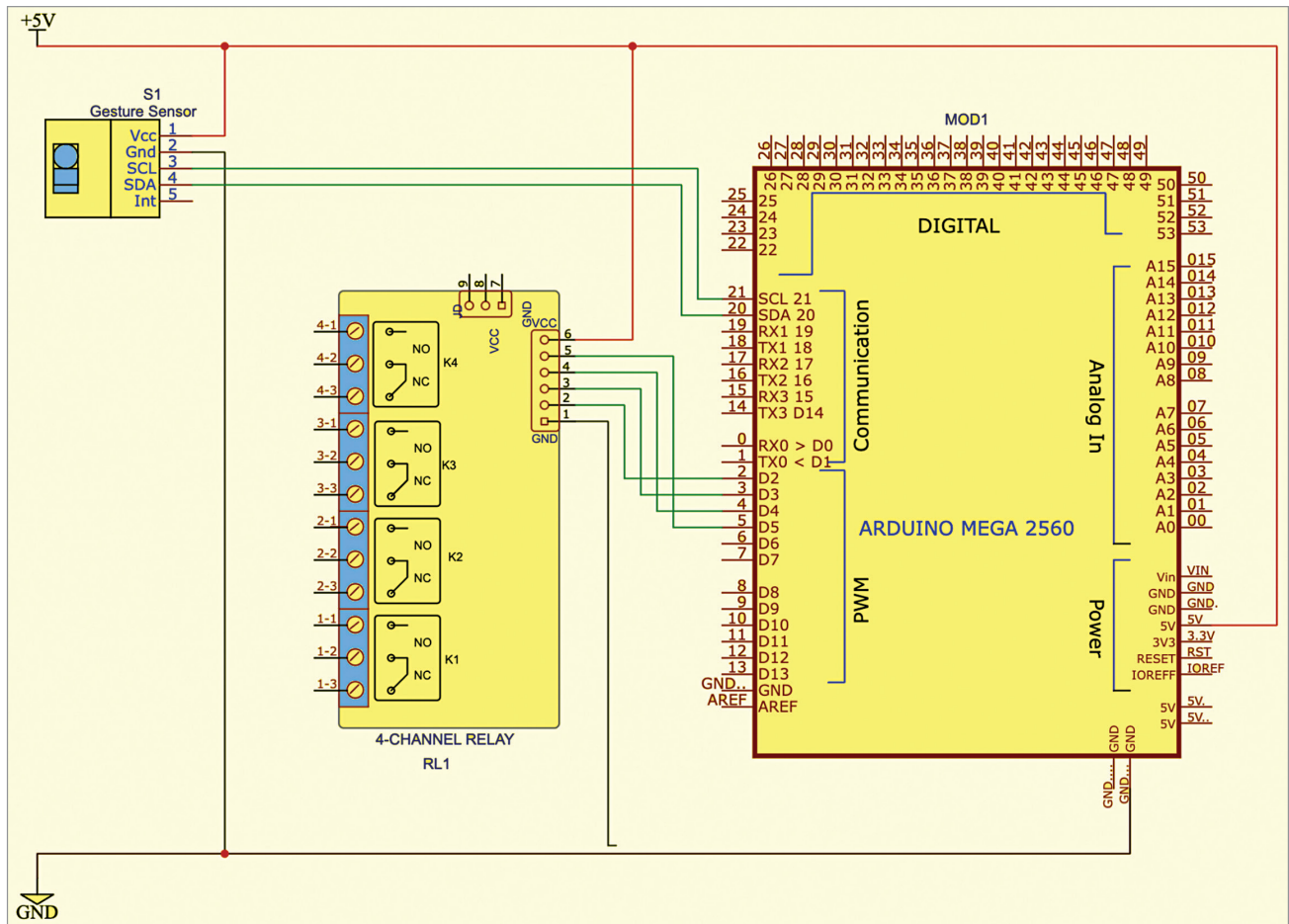


Fig. 2: Circuit diagram

connected to any available I/O pins on the board, with corresponding changes in the source code.

Software implementation

The system follows structured logic programmed in C++ using the Arduino IDE:

- **Gesture recognition.** Detects left, right, up, and down gestures
- **Appliance selection.** Uses a counter to track active relay positions from 1 to 4
- **Relay control.** Turns appliances ON or OFF based on up/down gestures
- **Circular navigation.** If the counter reaches 4 and a right gesture is detected, it resets to 1; similarly, a left gesture at 1 sets it back to 4

The code utilises the DFRobot gesture sensor library. The required libraries can be downloaded from the product's wiki page on DFRobot's official website: https://wiki.dfrobot.com/SKU_SEN0561_Gravity_GR10_30_Gesture_Sensor

Source code explanation

1. Initialising the gesture sensor.

```
DFRobot_GR10_30
gr10_30(GR10_30_DEVICE_ADDR,
&Wire);
```

The gesture sensor is initialised using the DFRobot GR10-30 library, which communicates via the I2C protocol.

2. Defining relay pins. Fig. 3 shows the code snippet used to define the relay pins. These pins are configured to connect the relay

module to the microcontroller.

3. Gesture-based appliance selection. This logic is responsible for incrementing or decrementing the currently selected appliance. It ensures cyclic navigation through four relay positions using left/right gestures. Fig. 4 shows the code snippet used to check and process gesture inputs.

4. Controlling appliances using gestures. Once an appliance is selected, up and down gestures are used to toggle its ON/OFF state. Left and right gestures help switch between appliances in a circular manner.

Construction and testing

Begin by downloading the complete source code from <https://sourcecode>.